

ORIGINAL RESEARCH

HEART FAILURE

Serum Magnesium, Outcomes, and the Effect of Empagliflozin in Heart Failure With Mildly Reduced and Preserved Ejection Fraction



Findings From EMPEROR-Preserved

João Pedro Ferreira, MD, PhD,^{a,b} Milton Packer, MD,^{c,d} Javed Butler, MD, MPH, MBA,^{e,f} Francisco Vasques-Nóvoa, MD,^a Pedro Marques, MD,^a Stuart Pocock, PhD,^g Gerasimos Filippatos, MD, PhD,^h Faiez Zannad, MD, PhD,^b Stefan D. Anker, MD, PhDⁱ

ABSTRACT

BACKGROUND Magnesium plays a central role in maintaining cellular homeostasis. Limited data exist on the clinical implications of magnesium derangements and the influence of sodium-glucose cotransporter 2 inhibitors on magnesium levels in heart failure with mildly reduced or preserved ejection fraction.

OBJECTIVES Using the EMPEROR-Preserved (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure with Preserved Ejection Fraction) population, the authors aimed to study the association of serum magnesium with outcomes, assess the impact of empagliflozin on serum magnesium levels, and explore the influence of serum magnesium on the effect of empagliflozin on the study outcomes.

METHODS Patients with heart failure with mildly reduced or preserved ejection fraction were randomized to receive placebo or empagliflozin 10 mg daily. Laboratory results were available at baseline; week 4; weeks 12, 32, and 52; and every 24 weeks thereafter. The median follow-up time was 26.2 months. The primary outcome was a composite of cardiovascular death or heart failure hospitalization.

RESULTS A total of 5,988 patients were included. The mean serum magnesium levels at baseline were 0.82 ± 0.11 mmol/L. The corresponding magnesium quintiles were as follows: Q1 = 0.67 ± 0.07 mmol/L (n = 1,291), Q2 = 0.78 ± 0.02 (n = 1,189), Q3 = 0.83 ± 0.01 (n = 1,378), Q4 = 0.88 ± 0.01 (n = 1,078), and Q5 = 0.96 ± 0.05 (n = 1,052). Patients with higher magnesium levels were older, had lower estimated glomerular filtration rate, and higher prevalence of atrial fibrillation. Conversely, patients with lower serum magnesium had diabetes and used thiazide-type diuretic agents more frequently. Placebo-treated patients experienced a higher risk of primary outcome events with higher magnesium levels (Q5). Empagliflozin (vs placebo) was associated with a more pronounced reduction of primary outcome events at higher baseline magnesium levels: Q1, HR: 1.05 [95% CI: 0.77-1.39]; Q2, HR: 0.85 [95% CI: 0.63-1.14]; Q3, HR: 0.88 [95% CI: 0.66-1.17]; Q4, HR: 0.62 [95% CI: 0.45-0.86]; Q5, HR: 0.60 [95% CI: 0.45-0.79]; interaction *P*-trend = 0.030. At week 4, empagliflozin had increased magnesium levels by 0.05 mmol/L, reaching a steady state thereafter.

CONCLUSIONS In EMPEROR-Preserved, higher serum magnesium levels were associated with a higher risk of primary outcome events among patients treated with placebo but not among patients treated with empagliflozin who experienced pronounced benefit. (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure with Preserved Ejection Fraction [EMPEROR-Preserved]; [NCT03057951](https://doi.org/10.1016/j.jchf.2025.102889)) (JACC Heart Fail. 2026;14:102889) © 2026 by the American College of Cardiology Foundation.

**ABBREVIATIONS
AND ACRONYMS****eGFR** = estimated glomerular filtration rate**HF** = heart failure**HFmrEF** = heart failure with mildly reduced ejection fraction**HFpEF** = heart failure with preserved ejection fraction**HFrEF** = heart failure with reduced ejection fraction**LVEF** = left ventricular ejection fraction**SGLT2** = sodium-glucose cotransporter 2**T2D** = type 2 diabetes mellitus

Magnesium is an abundant intracellular cation essential for all energy-dependent cellular processes.^{1,2} In humans, most magnesium is stored in bone, soft tissues, and muscle, whereas the ~1% that circulates extracellularly is the fraction measured in serum with normal serum magnesium concentration in adults ranging between 0.7 and 1.0 mmol/L (1.7 to 2.4 mg/dL).³ Magnesium is essential for heart contractility and rhythm, as well as for endothelial and vascular function.^{3,4} Despite its importance, clinical data on the role of magnesium in heart failure (HF) are scarce.⁵⁻⁷

Most studies assessing the prognostic role of magnesium in chronic HF were performed in patients with heart failure and reduced ejection fraction (HFrEF) in whom the presence hypomagnesemia has been associated with an increased risk of cardiovascular mortality, including sudden cardiac death.⁸ In a prior report from EMPEROR-Reduced (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction; [NCT03057977](#)) including outpatients with HFrEF, we found that the presence of type 2 diabetes (T2D) was associated with lower serum magnesium levels, and that low serum magnesium (≤ 0.75 mg/dL) at baseline was modestly associated with higher rates of the composite of HF hospitalization or cardiovascular death and adverse kidney outcomes, without observed increase in risk at higher magnesium levels. Confirming data from prior reports,⁹⁻¹⁴ treatment with empagliflozin rapidly increased serum magnesium levels. Interestingly, in EMPEROR-Reduced the response to empagliflozin (vs placebo) appeared to be more pronounced among patients presenting with lower serum magnesium at baseline. These findings highlight the importance of measuring serum magnesium in HFrEF and inform about the effects of empagliflozin on magnesium

levels of these patients, as well as the expected treatment response according to magnesium levels.

Information on the implications of serum magnesium levels in patients with heart failure with mildly reduced ejection fraction (HFmrEF) and heart failure with preserved ejection fraction (HFpEF) is very limited.¹⁵ The effect of sodium-glucose cotransporter 2 (SGLT2) inhibitors on serum magnesium levels and the influence of magnesium levels on the cardiovascular and renal effects of SGLT2 inhibitors in patients with HFmrEF/HFpEF is currently unknown.

Using the EMPEROR-Preserved (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure with Preserved Ejection Fraction; [NCT03057951](#)) population we aim to: 1) describe the patient's characteristics across different levels of serum magnesium; 2) study the association of serum magnesium with cardiovascular and kidney outcomes; 3) assess the impact of empagliflozin on serum magnesium levels over time; and 4) explore the potential influence of serum magnesium at baseline on the effect of empagliflozin on the study outcomes.

METHODS

EMPEROR-Preserved was a randomized, double-blind, parallel-group, placebo-controlled and event-driven trial, whose design and primary results have been described previously.^{16,17} Participants were adults with chronic HF with NYHA functional class II-IV symptoms for at least 3 months and a left ventricular ejection fraction (LVEF) $>40\%$ with no prior measurement $\leq 40\%$ under stable conditions. Patients were required to have elevated N-terminal pro-hormone B-type natriuretic peptide levels (>900 pg/mL or >300 pg/mL in patients with or without atrial fibrillation, respectively) and have evidence of structural heart disease (left ventricular hypertrophy or increased left atrial size) or a documented HF hospitalization within 12 months. The protocol was approved by the Ethical Committee of

From the ^aRISE-Health, Department of Surgery and Physiology, Faculty of Medicine, University of Porto, Porto, Portugal; ^bUniversité de Lorraine, Inserm, Centre d'Investigations Cliniques, - Plurithématique 14-33, and Inserm U1116, CHRU Nancy, F-CRIN INI-CRCT (Cardiovascular and Renal Clinical Trialists), Nancy, France; ^cBaylor Heart and Vascular Institute, Baylor University Medical Center, Dallas, Texas, USA; ^dImperial College, London, United Kingdom; ^eBaylor Scott and White Research Institute, Dallas, Texas, USA; ^fUniversity of Mississippi, Jackson, Mississippi, USA; ^gDepartment of Medical Statistics, London School of Hygiene and Tropical Medicine, London, United Kingdom; ^hNational and Kapodistrian University of Athens, School of Medicine, Attikon University Hospital, Athens, Greece; and the ⁱDepartment of Cardiology (CVK) of German Heart Center Charité, German Centre for Cardiovascular Research (DZHK) partner site Berlin, Charité Universitätsmedizin, Berlin, Germany. Biykem Bozkurt, MD, PhD, served as acting Editor-in-Chief and main adjudicator for this paper.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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each of the 622 participating sites in 23 countries. All patients gave written informed consent to participate in the study.

STUDY VISITS. Patients were randomized in a double-blind manner to receive placebo or empagliflozin 10 mg daily (in a 1:1 ratio), in addition to their usual therapy. Patients were periodically assessed at study visits for laboratory results, including serum magnesium, which was available at baseline; week 4; weeks 12, 32, and 52; and every 24 weeks thereafter until the end of the study. All randomized patients were followed for the occurrence of outcomes for the entire duration of the trial. The median duration of follow-up for the primary outcome was 26.2 months (Q1-Q3: 18.1-33.1 months).

STUDY ENDPOINTS. The primary endpoint was the composite of adjudicated cardiovascular death or hospitalization for HF, analyzed as time to first event. In addition, we also analyzed the individual components of the primary outcome, sudden cardiac death, all-cause mortality, and a composite kidney outcome defined by a sustained $\geq 40\%$ drop in estimated glomerular filtration rate (eGFR), a sustained eGFR < 15 mL/min/1.73 m², or the need for chronic dialysis or kidney transplantation.

STATISTICAL ANALYSIS. Baseline serum magnesium levels were divided in quintiles. Patients' characteristics were compared across serum magnesium levels using the appropriate trend tests. The associations of serum magnesium levels with the study outcomes were assessed in the placebo group with the third category (Q3) as referent, using Cox proportional hazards models adjusted for the prespecified baseline covariates of age (continuous), sex (men vs women), race (White vs other; used instead of geographical region to be uniformized with EMPEROR-Reduced), diabetes status (yes vs no), LVEF (continuous), and eGFR (continuous). Further adjustment was performed including all the variables with statistically significant differences across serum magnesium levels as reported in [Table 1](#). Proportional hazard assumption was tested based on Schoenfeld residuals after fitting a Cox model. No evidence of proportional hazard assumption violation was found. For the analysis of changes in serum magnesium levels over time, treatment effects were assessed based on changes from baseline using a mixed model for repeated measures adjusted for the preceding referenced covariates plus the baseline magnesium value. Independent predictors of serum magnesium increase from baseline to week 12 were studied using a stepwise backward model with a value of $P < 0.05$ set for a variable to be retained in the final model, all

variables with a value of $P < 0.05$ from [Table 1](#) plus the randomized treatment were inserted in the full initial model. Differences in the effect of empagliflozin across serum magnesium levels were assessed by interaction terms with an interaction P value for trend across serum magnesium Qs. A total of 75 patients (of 5,988) had missing serum magnesium levels at baseline, and in these 75 (1.3%) cases the missing values were imputed with the median serum magnesium value of the cohort. P values and 95% CIs presented in this report have not been adjusted for multiplicity given the exploratory nature of this work. All analyses were performed using STATA (StataCorp LLC, Stata Statistical Software: Release 19) using the virtual environment at Vivli.

RESULTS

PATIENTS' CHARACTERISTICS ACROSS SERUM MAGNESIUM LEVELS.

A total of 5,988 patients were included in EMPEROR-Preserved. The mean serum magnesium levels at baseline in the entire cohort were 0.82 ± 0.11 mmol/L. The corresponding serum magnesium values across quintiles were: Q1: 0.67 ± 0.07 mmol/L ($n = 1,291$); Q2: 0.78 ± 0.02 ($n = 1,189$); Q3: 0.83 ± 0.01 ($n = 1,378$); Q4: 0.88 ± 0.01 ($n = 1,078$); and Q5: 0.96 ± 0.05 ($n = 1,052$). Q1 started at serum magnesium levels < 0.75 mmol/L and the minimum serum magnesium value was 0.28 mmol/L. Q5 started at serum magnesium levels > 0.90 mmol/L and the maximum serum magnesium value was 1.44 mmol/L. At higher serum magnesium levels patients were older, had more frequent atrial fibrillation or flutter, higher N-terminal pro-B-type natriuretic peptide levels, and had lower eGFR, lower systolic blood pressure, and less frequent hypertension diagnosis, slower heart rate, and lower body mass index; they also used loop diuretic agents more frequently and thiazide-type diuretic agents less frequently (all P -trend < 0.001). The prevalence of T2D decreased gradually as serum magnesium levels increased (P -trend < 0.001). No differences across serum magnesium levels were observed regarding serum calcium values. Despite statistical differences, the values of serum sodium, serum potassium, and albumin were similar across serum magnesium levels with absolute differences between Q1 and Q5 not > 1 mmol/L, 0.1 mmol/L, and 0.1 g/dL, respectively ([Table 1](#)).

PLACEBO-GROUP OUTCOME ASSOCIATIONS OF SERUM MAGNESIUM.

Compared with intermediate serum magnesium levels (Q3), patients with higher levels (Q5) presented a higher risk of experiencing primary outcome events: prespecified covariate adjusted HR: 1.59 [95% CI: 1.21-2.08]; $P = 0.001$ and

TABLE 1 Patients' Characteristics by Quintiles of Serum Magnesium (N = 5,988)

	Q1 (0.28-0.74 mmol/L) (n = 1,291)	Q2 (0.75-0.80 mmol/L) (n = 1,189)	Q3 (0.81-0.85 mmol/L) (n = 1,378)	Q4 (0.86-0.90 mmol/L) (n = 1,078)	Q5 (0.91-1.44 mmol/L) (n = 1,052)	P Trend
Mg ²⁺ , mmol/L	0.67 ± 0.07	0.78 ± 0.02	0.83 ± 0.01	0.88 ± 0.01	0.96 ± 0.05	<0.001
Age, y	70.8 ± 9.1	70.9 ± 9.7	71.7 ± 9.8	72.5 ± 9.7	74.2 ± 9.1	<0.001
Men	723 (56.0)	671 (56.4)	783 (56.8)	568 (52.7)	567 (53.9)	0.20
White	1,040 (80.6)	904 (76.0)	1,019 (73.9)	816 (75.7)	763 (72.5)	<0.001
Hypertension	1,210 (93.7)	1,088 (91.5)	1,228 (89.1)	961 (89.2)	937 (89.1)	<0.001
Diabetes	917 (71.0)	645 (54.3)	577 (41.9)	380 (35.3)	394 (37.5)	<0.001
AF/flutter	627 (48.6)	606 (51.0)	699 (50.7)	605 (56.1)	607 (57.7)	<0.001
HFH past 12 mo	299 (23.2)	276 (23.2)	321 (23.3)	264 (24.5)	308 (29.3)	0.003
Ischemic HF	520 (40.3)	430 (36.2)	485 (35.2)	359 (33.3)	323 (30.7)	<0.001
NYHA functional class III/IV	241 (18.7)	216 (18.2)	223 (16.2)	187 (17.4)	234 (22.2)	0.003
Peripheral edema	414 (32.1)	346 (29.1)	386 (28.0)	277 (25.7)	324 (30.8)	0.008
SBP, mm Hg	133 ± 16	132 ± 15	132 ± 16	131 ± 16	131 ± 16	<0.001
Heart rate	72 ± 12	70 ± 12	70 ± 12	70 ± 12	69 ± 12	<0.001
BMI, kg/m ²	30.9 ± 5.9	30.1 ± 5.9	29.6 ± 5.6	29.3 ± 5.8	29.1 ± 5.9	<0.001
LVEF, %	54 ± 9	54 ± 9	54 ± 9	54 ± 9	55 ± 9	0.003
Hemoglobin, g/dL	13.1 ± 1.6	13.5 ± 1.6	13.4 ± 1.4	13.4 ± 1.6	13.2 ± 1.7	<0.001
eGFR, mL/min	63.1 ± 19.9	64.1 ± 19.8	63.0 ± 19.6	58.1 ± 18.7	53.1 ± 19.0	<0.001
eGFR <60 mL/min	583 (45.2)	516 (43.4)	619 (44.9)	591 (54.8)	679 (64.5)	<0.001
UACR, mg/g ^a	25 (9-94)	22 (7-83)	19 (7-68)	19 (7-49)	21 (8-67)	<0.001
NT-proBNP, pg/mL ^a	893 (486-1,605)	896 (896-1,630)	933 (491-1,628)	1,052 (505-1,776)	1,178 (561-2,085)	<0.001
Serum calcium, mmol/L	2.4 ± 0.1	2.3 ± 0.1	2.3 ± 0.1	2.4 ± 0.1	2.3 ± 0.1	0.37
Serum sodium, mmol/L	141 ± 3	141 ± 3	141 ± 3	141 ± 3	141 ± 3	<0.001
Serum potassium, mmol/L	4.5 ± 0.5	4.5 ± 0.5	4.5 ± 0.4	4.5 ± 0.5	4.6 ± 0.5	<0.001
Albumin, g/dL	4.3 ± 0.3	4.3 ± 0.3	4.4 ± 0.3	4.4 ± 0.3	4.4 ± 0.3	<0.001
Loop diuretic	797 (61.7)	764 (64.3)	880 (63.9)	770 (71.4)	843 (80.1)	<0.001
ACEI/ARB	1,078 (83.5)	964 (81.1)	1,079 (78.3)	826 (76.6)	758 (72.1)	<0.001
Beta-blocker	1,147 (88.9)	1,035 (87.1)	1,177 (85.4)	931 (86.4)	877 (83.4)	0.003
MRA	490 (38.0)	461 (38.8)	499 (36.2)	404 (37.5)	390 (37.1)	0.74
Thiazide diuretic	371 (28.7)	245 (20.6)	269 (19.5)	191 (17.7)	163 (15.5)	<0.001

Values are mean ± SD, n (%), or median (Q1-Q3), unless otherwise indicated. ^aMedian (percentile 25th to 27th).

ACEI = angiotensin-converting enzyme inhibitor; AF = atrial fibrillation; ARB = angiotensin receptor blocker; BMI = body mass index; eGFR = estimated glomerular filtration rate; HFH = heart failure hospitalization; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide; SBP = systolic blood pressure; UACR = urinary albumin-to-creatinine ratio.

fully adjusted HR: 1.44 [95% CI: 1.09-1.90]; $P = 0.010$; whereas those with lower serum magnesium levels (Q1) presented a similar risk: pre-specified covariate adjusted HR: 0.96 [95% CI: 0.73-1.27]; $P = 0.79$ and fully adjusted HR: 0.93 [95% CI: 0.70-1.24]; $P = 0.62$. The association of higher serum magnesium with higher event rates levels was driven both by an increased risk of cardiovascular death and HF hospitalizations: pre-specified covariate adjusted HR: 1.54 [95% CI: 1.01-2.34] and HR: 1.55 [95% CI: 1.14-2.11], respectively. The risk of adverse kidney outcomes, sudden death, or all-cause mortality did not significantly differ by serum magnesium levels. These associations were largely maintained after covariate adjustment for all the variables with a value of $P < 0.05$ in [Tables 1 and 2](#) and [Figure 1](#).

Outcome associations limited to patients with HFpEF (LVEF $\geq 50\%$) overlap those presented for the

entire cohort (magnesium-by-LVEF interaction $P > 0.10$ for all outcomes) ([Supplemental Table 1](#)).

TREATMENT EFFECT OF EMPAGLIFLOZIN VS PLACEBO ACROSS SERUM MAGNESIUM LEVELS.

The effect of empagliflozin (vs placebo) on the study primary outcome appeared to be more pronounced at higher baseline serum magnesium levels: Q1: HR: 1.05 [95% CI: 0.77-1.39]; Q2: HR: 0.85 [95% CI: 0.63-1.14]; Q3: HR: 0.88 [95% CI: 0.66-1.17]; Q4: HR: 0.62 [95% CI: 0.45-0.86]; Q5: HR: 0.60 [95% CI: 0.45-0.79]; interaction P -trend = 0.030 (and treatment-by-magnesium as continuous variable interaction $P = 0.012$). A similar pattern of an apparent pronounced treatment benefit at higher serum magnesium levels was observed both for cardiovascular death and HF hospitalizations, despite a nonsignificant statistical interaction for the individual components of the primary outcome likely due to low statistical power. The effect of

TABLE 2 Placebo-Group Outcome Associations by Quintiles of Serum Magnesium (N = 2,991)

	Q1 (0.28-0.74 mmol/L) (N = 673)	Q2 (0.75-0.80 mmol/L) (N = 617)	Q3 (0.81-0.85 mmol/L) (N = 670)	Q4 (0.86-0.90 mmol/L) (N = 524)	Q5 (0.91-1.44 mmol/L) (N = 507)
Cardiovascular death or HFH					
Events, n/N (%)	102 (15.2)	97 (15.7)	98 (14.6)	92 (17.6)	122 (24.1)
HR (95% CI)	1.02 (0.78-1.35)	1.08 (0.81-1.42)	Ref. (1)	1.20 (0.90-1.59)	1.76 (1.34-2.30)
P value	0.87	0.61		0.21	<0.001
Adj. HR (95% CI) ¹	0.96 (0.73-1.27)	1.06 (0.80-1.40)	Ref. (1)	1.16 (0.87-1.55)	1.59 (1.21-2.08)
P value	0.79	0.70		0.30	0.001
Adj. HR (95% CI) ²	0.93 (0.70-1.24)	1.0 (0.75-1.33)	Ref. (1)	1.19 (0.89-1.59)	1.44 (1.09-1.90)
P value	0.62	0.99		0.24	0.010
Cardiovascular death					
Events, n/N (%)	61 (9.1)	51 (8.3)	40 (6.0)	42 (8.0)	50 (9.9)
HR (95% CI)	1.53 (1.02-2.27)	1.40 (0.93-2.12)	Ref. (1)	1.33 (0.86-2.05)	1.72 (1.13-2.60)
P value	0.038	0.11		0.20	0.011
Adj. HR (95% CI) ¹	1.57 (1.04-2.35)	1.42 (0.94-2.15)	Ref. (1)	1.25 (0.81-1.93)	1.54 (1.01-2.34)
P value	0.030	0.10		0.32	0.045
Adj. HR (95% CI) ²	1.50 (0.99-2.27)	1.36 (0.89-2.07)	Ref. (1)	1.34 (0.86-2.08)	1.56 (1.01-2.40)
P value	0.052	0.15		0.19	0.043
HFH					
Events, n/N (%)	58 (8.6)	61 (9.9)	75 (11.2)	65 (12.4)	93 (18.3)
HR (95% CI)	0.76 (0.54-1.07)	0.88 (0.63-1.23)	Ref. (1)	1.11 (0.79-1.55)	1.75 (1.29-2.37)
P value	0.12	0.47		0.53	<0.001
Adj. HR (95% CI) ¹	0.69 (0.48-0.97)	0.85 (0.61-1.20)	Ref. (1)	1.08 (0.77-1.51)	1.55 (1.14-2.11)
P value	0.033			0.66	0.005
Adj. HR (95% CI) ²	0.66 (0.47-0.94)	0.80 (0.57-1.12)	Ref. (1)	1.05 (0.75-1.48)	1.28 (0.93-1.76)
P value	0.021	0.20		0.76	0.13
Kidney endpoint					
Events, n/N (%)	29 (4.3)	25 (4.1)	26 (3.9)	16 (3.1)	16 (3.2)
HR (95% CI)	1.11 (0.65-1.88)	1.06 (0.61-1.83)	Ref. (1)	0.78 (0.42-1.46)	0.85 (0.46-1.59)
P value	0.70	0.84		0.44	0.61
Adj. HR (95% CI) ¹	1.03 (0.60-1.76)	1.05 (0.60-1.82)	Ref. (1)	0.76 (0.41-1.43)	0.77 (0.41-1.44)
P value	0.92	0.86		0.40	0.41
Adj. HR (95% CI) ²	0.98 (0.57-1.70)	1.02 (0.58-1.78)	Ref. (1)	0.77 (0.41-1.45)	0.69 (0.36-1.33)
P value	0.94	0.95		0.41	0.27
Sudden cardiac death					
Events, n/N (%)	28 (4.2)	25 (4.1)	19 (2.8)	21 (4.0)	25 (4.9)
HR (95% CI)	1.48 (0.82-2.65)	1.45 (0.80-2.63)	Ref. (1)	1.39 (0.75-2.58)	1.81 (1.00-3.29)
P value	0.19	0.22		0.30	0.051
Adj. HR (95% CI) ¹	1.56 (0.87-2.83)	1.47 (0.81-2.68)	Ref. (1)	1.31 (0.70-2.45)	1.65 (0.90-3.03)
P value	0.14	0.21		0.39	0.10
Adj. HR (95% CI) ²	1.59 (0.87-2.90)	1.39 (0.76-2.55)	Ref. (1)	1.36 (0.72-2.55)	1.64 (0.88-3.04)
P value	0.13	0.28		0.34	0.12
All-cause mortality					
Events, n/N (%)	101 (15.0)	93 (15.1)	84 (12.5)	70 (13.4)	79 (15.6)
HR (95% CI)	1.20 (0.90-1.60)	1.22 (0.91-1.64)	Ref. (1)	1.05 (0.77-1.45)	1.30 (0.95-1.76)
P value	0.22	0.18		0.75	0.097
Adj. HR (95% CI) ¹	1.22 (0.91-1.63)	1.24 (0.92-1.67)	Ref. (1)	0.97 (0.71-1.34)	1.12 (0.82-1.53)
P value	0.19	0.16		0.86	0.48
Adj. HR (95% CI) ²	1.19 (0.88-1.60)	1.16 (0.86-1.57)	Ref. (1)	1.02 (0.74-1.41)	1.10 (0.80-1.51)
P value	0.26	0.32		0.89	0.55

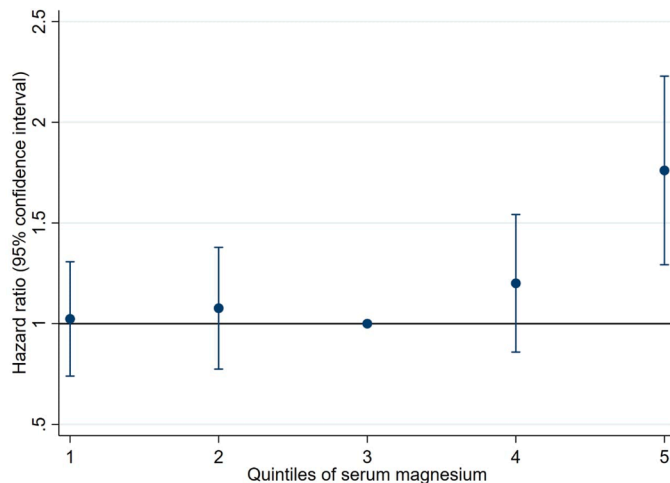
Adj.¹ = model adjusted for the prespecified variables of age, sex, race, diabetes, eGFR, and LVEF; Adj.² = model adjusted for all the statistically significant differences found in [Table 1](#); Kidney endpoint = a sustained reduction in eGFR, defined as a decline of $\geq 40\%$. From baseline or sustained eGFR < 15 mL/min/1.73 m² (for patients with baseline eGFR ≥ 30) or sustained eGFR < 10 (for patients with baseline eGFR < 30), initiation of dialysis, kidney transplantation, or kidney-related death.
Ref. (1) = referent category (HR = 1); other Abbreviations as in [Table 1](#).

empagliflozin on kidney endpoint, sudden death, and all-cause mortality was not modified by baseline serum magnesium levels (all interaction P trend > 0.10) ([Table 3](#), [Figure 2](#)).

The effect of empagliflozin (vs placebo) on the study outcomes among patients with HFpEF

(LVEF $\geq 50\%$) overlap those presented for the entire cohort (magnesium-by-treatment-by-LVEF interaction $P > 0.1$ for all outcomes) ([Supplemental Table 2](#)).

INFLUENCE OF EMPAGLIFLOZIN VS PLACEBO ON SERUM MAGNESIUM LEVELS. Throughout the follow-up, compared with placebo, empagliflozin increased

FIGURE 1 Association Between Quintiles of Serum Mg^{2+} Levels and the Study Primary Outcome in the Placebo Group

The minimum and maximum values of serum magnesium (mmol/L) in each quintile are: Q1: 0.28-0.74 (n = 673); Q2: 0.75-0.80 (n = 617); Q3: 0.81-0.85 (n = 670); Q4: 0.86-0.90 (n = 524); and Q5: 0.91-1.44 (n = 507).

serum magnesium levels by 0.05 mmol/L on average, with rapid increase in magnesium levels observed as early as week 4, reaching a steady state thereafter (Figures 3A and 3B). The odds of experiencing low magnesium levels, here defined as serum magnesium levels <0.75 mmol/L (Q1), were reduced with empagliflozin: OR: 0.31 [95% CI: 0.26-0.36]; $P < 0.001$; whereas the odds of experiencing high magnesium levels, here defined as serum magnesium levels >0.90 mmol/L (Q5), were increased with empagliflozin: OR: 3.34 [95% CI: 2.93-3.81]; $P < 0.001$ (Figures 3C and 3D).

The correlations between changes from baseline to week 12 in serum magnesium levels and the corresponding changes in hemoglobin, hematocrit, and serum potassium levels were weak (Spearman $Rho \leq 0.2$ for all) and are presented in Supplemental Table 3.

The strongest independent predictor of serum magnesium increase from baseline to week 12 was randomization to empagliflozin (vs placebo) (Supplemental Table 4). Empagliflozin (vs placebo) increased serum magnesium levels over time regardless of the baseline magnesium levels; still, the increase was slightly more pronounced among patients with lower magnesium at baseline (Q1: 0.06 mmol/L) vs the remainder of the categories (0.04 to 0.05 mmol/L) (Supplemental Table 5).

The main findings of this study are summarized in the Central Illustration.

DISCUSSION

To the best of our knowledge this is the first large study to explore the associations of serum magnesium with cardiovascular and renal outcomes in HFmrEF/HFpEF. Moreover, we study how magnesium levels may influence the effect of empagliflozin and how empagliflozin affects magnesium levels over time. Several original and clinically relevant findings should be highlighted: 1) patients with higher serum magnesium levels were older, had a higher prevalence of atrial fibrillation, lower eGFR, and used loop diuretic agents more frequently, whereas the presence of T2D and the use of thiazide-type diuretic agents was associated with lower serum magnesium levels; 2) higher serum magnesium (ie, Q5 >0.90 mmol/L) levels were associated with poorer cardiovascular outcomes if on placebo; 3) the effect of empagliflozin (vs placebo) appeared to be influenced by baseline serum magnesium, with a more pronounced benefit observed among patients with higher serum magnesium levels at baseline; and 4) empagliflozin increased serum magnesium by 0.05 mmol/L on average, with a nearly 70% relative reduction in the odds of experiencing low serum magnesium (<0.75 mmol/L); conversely, the odds of experiencing high serum magnesium levels (>0.90 mmol/L) increased more than 3-fold with empagliflozin. Notwithstanding, it should be noted that our extreme quintiles (Q1 and Q5) included patients considered to have hypomagnesemia and hypermagnesemia, respectively, but these categories also include some patients with magnesium levels that would fit into the “normal” range of 0.7 to 1.0 mmol/L.⁵ The increase of serum magnesium with empagliflozin was rapid (observed as early as 4 weeks) and reached a steady state thereafter.⁹ These findings may have clinical implications for patients with HFmrEF/HFpEF, as they suggest an association between higher magnesium levels and an increased risk of primary outcome events under placebo, a risk that seems to be reversed with empagliflozin treatment.

Both hypomagnesemia and hypermagnesemia have previously been associated with worse outcomes in chronic HF, with most studies being conducted in patients with HFrEF.¹⁸⁻²¹ Although it remains unclear whether magnesium is a direct mediator of HF progression or merely a marker of increased risk reflecting important comorbidities (eg, older age and renal dysfunction), several mechanistic hypotheses support a potentially detrimental role of elevated magnesium levels in HFmrEF/HFpEF, namely: 1) at the cardiomyocyte level, magnesium competes with calcium for key

TABLE 3 Treatment Effect (Empagliflozin vs Placebo) by Quintiles of Serum Magnesium (N = 5,988)

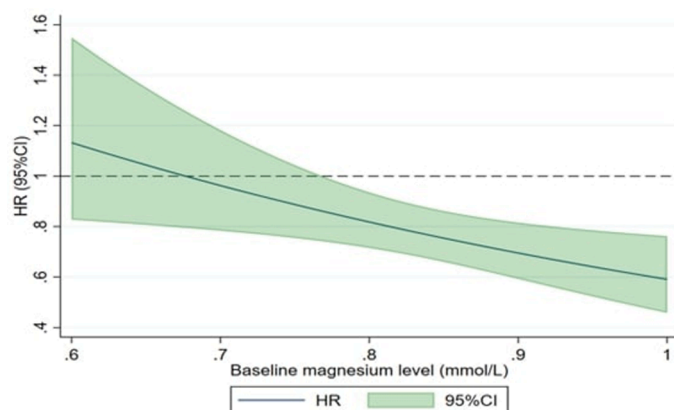
	Q1 (0.28-0.74 mmol/L) (n = 1,291)		Q2 (0.75-0.80 mmol/L) (n = 1,189)		Q3 (0.81-0.85 mmol/L) (n = 1,378)		Q4 (0.86-0.90 mmol/L) (n = 1,078)		Q5 (0.91-1.44 mmol/L) (n = 1,052)	
	Pbo.	Empa.	Pbo.	Empa.	Pbo.	Empa.	Pbo.	Empa.	Pbo.	Empa.
Cardiovascular death or HFH (treatment-by-Mg ²⁺ interaction P trend = 0.030)										
Events, n/N (%)	102/673 (15.2)	95/618 (15.4)	97/617 (15.7)	77/572 (13.5)	98/670 (14.6)	93/708 (13.1)	92/524 (17.6)	64/554 (11.6)	122/507 (24.1)	86/545 (15.8)
Empa. vs Pbo. HR (95% CI)	1.05 (0.77-1.39)		0.85 (0.63-1.14)		0.88 (0.66-1.17)		0.62 (0.45-0.86)		0.60 (0.45-0.79)	
Cardiovascular death (treatment-by-Mg ²⁺ interaction P trend = 0.25)										
Events, n/N (%)	61/673 (9.1)	53/618 (8.6)	51/617 (8.3)	48/572 (8.4)	40/670 (6.0)	49/708 (6.9)	42/524 (8.0)	31/554 (5.6)	50/507 (9.9)	38/545 (7.0)
Empa. vs Pbo. HR (95% CI)	0.99 (0.68-1.42)		1.03 (0.69-1.53)		1.17 (0.77-1.77)		0.68 (0.43-1.09)		0.67 (0.44-1.02)	
HFH (treatment-by-Mg ²⁺ interaction P trend = 0.29)										
Events, n/N (%)	58/673 (8.6)	52/618 (8.4)	61/617 (9.9)	43/572 (7.5)	75/670 (11.2)	53/708 (7.5)	65/524 (12.4)	45/554 (8.1)	93/507 (18.3)	66/545 (12.1)
Empa. vs Pbo. HR (95%CI)	1.01 (0.69-1.47)		0.75 (0.51-1.11)		0.66 (0.46-0.93)		0.62 (0.43-0.91)		0.61 (0.44-0.83)	
Kidney endpoint (treatment-by-Mg ²⁺ interaction P trend = 0.40)										
Events, n/N (%)	29/673 (4.3)	27/618 (4.4)	25/617 (4.1)	29/572 (5.1)	26/670 (3.9)	16/708 (2.3)	16/524 (3.1)	20/554 (3.6)	16/507 (3.2)	16/545 (2.9)
Empa. vs Pbo. HR (95% CI)	1.07 (0.63-1.80)		1.27 (0.74-2.16)		0.58 (0.31-1.09)		1.15 (0.60-2.23)		0.87 (0.43-1.73)	
Sudden cardiac death (treatment-by-Mg ²⁺ interaction P trend = 0.22)										
Events, n/N (%)	28/673 (4.2)	27/618 (4.4)	25/617 (4.1)	21/572 (3.7)	19/670 (2.8)	26/708 (3.7)	21/524 (4.0)	12/554 (2.2)	25/507 (4.9)	17/545 (3.1)
Empa. vs Pbo. HR (95% CI)	1.09 (0.64-1.85)		0.92 (0.52-1.65)		1.30 (0.72-2.35)		0.53 (0.26-1.08)		0.60 (0.32-1.11)	
All-cause mortality (treatment-by-Mg ²⁺ interaction P trend = 0.19)										
Events, n/N (%)	101/673 (15.0)	109/618 (17.6)	93/617 (15.1)	78/572 (13.6)	84/670 (12.5)	97/708 (13.7)	70/524 (13.4)	70/554 (12.6)	79/507 (15.6)	68/545 (12.5)
Empa. vs Pbo. HR (95% CI)	1.23 (0.94-1.62)		0.92 (0.68-1.24)		1.10 (0.82-1.48)		0.93 (0.66-1.29)		0.75 (0.55-1.04)	

Empa. = empagliflozin; Pbo. = placebo; other abbreviations as in Table 1.

calcium-handling proteins (eg, RyR2, SERCA2a, and NCX), potentially impairing calcium cycling, delaying relaxation, and contributing to elevated ventricular filling pressures;^{22,23} 2) magnesium inhibits norepinephrine release from sympathetic nerve terminals by blocking presynaptic voltage-gated Ca²⁺ channels, leading to reduced sympathetic tone, negative chronotropic effects, and slowed electrical conduction (eg, atrioventricular node), which may compromise cardiac performance, particularly under stress or exertion.²⁴

Interestingly, our findings contrast with prior observations in patients with HFrEF who participated in the EMPEROR-Reduced trial in whom low (but not high) magnesium levels were associated with poor HF and kidney outcomes, and the benefits of empagliflozin appeared more pronounced among patients with low baseline serum magnesium levels. It is possible that in HFmrEF/HFpEF the placebo-group associations of high serum magnesium levels with outcomes are confounded by impaired kidney function, age, and atrial fibrillation all more prevalent

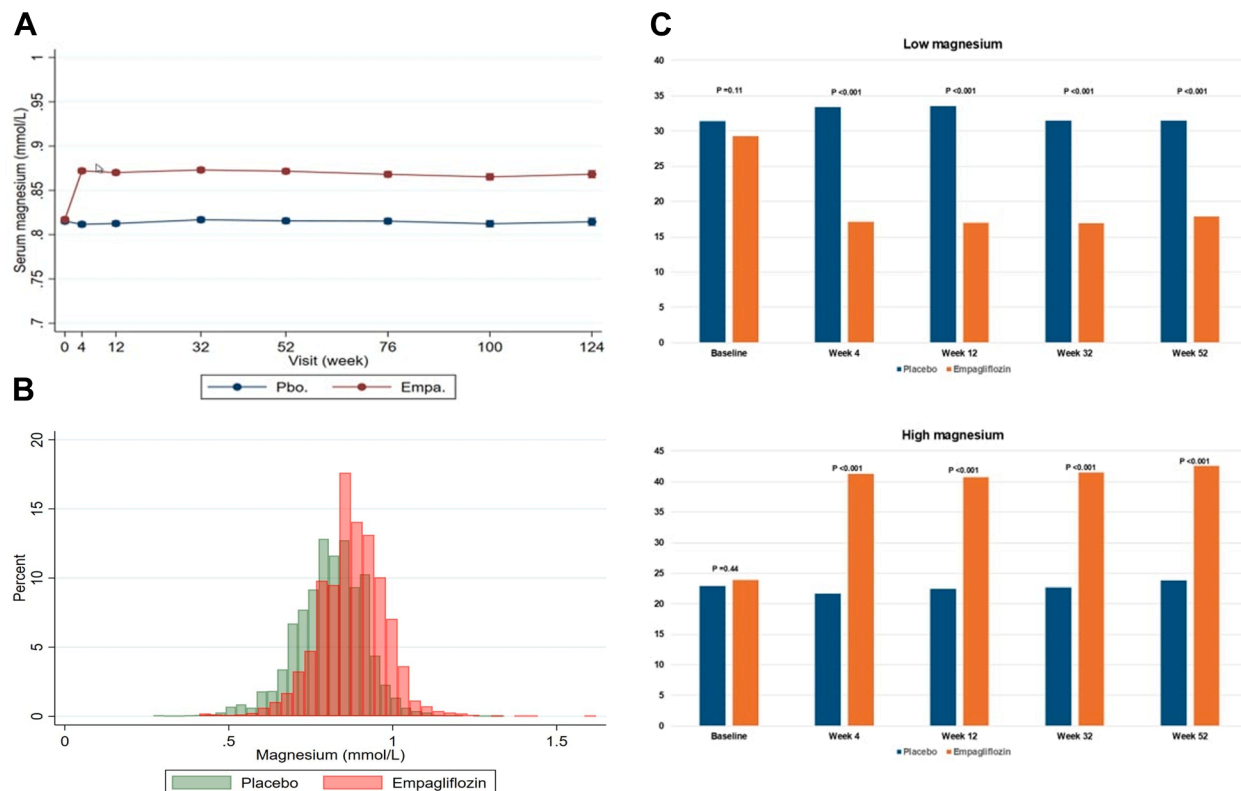
FIGURE 2 Treatment Effect of Empagliflozin vs Placebo for the Primary Outcome Across Quintiles of Serum Mg^{2+} Quintiles



The treatment effect appeared more pronounced at higher magnesium levels.
Treatment-by- Mg^{2+} interaction $P = 0.012$.

among patients with higher serum magnesium levels and all associated with poor outcomes in EMPEROR-Preserved.²⁵⁻²⁷ The correlation of high magnesium levels and HF outcomes in HFmrEF/HFpEF have been poorly documented.^{8,15} Therefore, we cannot ascertain whether we are observing a true difference between HFmrEF/HFpEF and HFREF or if we are being confounded by the higher prevalence of comorbid conditions in HFmrEF/HFpEF compared with HFREF; still, the associations of higher serum magnesium and poorer cardiovascular outcomes in HFmrEF/HFpEF remained significant even after extensive covariate adjustment suggesting a possible direct influence of high serum magnesium on adverse HF outcomes among placebo-treated patients. In either case, having higher serum magnesium levels does not seem to be problematic in HFmrEF/HFpEF as long as patients are treated with an SGLT2 inhibitor. This also raises the question of why the magnitude of benefit with

FIGURE 3 Influence of Empagliflozin on the Levels of Serum Mg^{2+}



Low magnesium, defined as a serum $Mg^{2+} \leq 0.74$ mmol/L (ie, Q1); high magnesium, defined as a serum $Mg^{2+} \geq 0.91$ mmol/L (ie, Q5). (A) On average empagliflozin increased serum Mg^{2+} by 0.047 [95% CI: 0.044-0.049 mmol/L]; $P < 0.001$. (B) Serum magnesium levels were shifted toward the right with empagliflozin. (C) The odds of experiencing hypomagnesemia were reduced with empagliflozin: OR: 0.31 [95% CI: 0.26-0.36]; $P < 0.001$, and the odds of experiencing hypermagnesemia were increased with empagliflozin: OR: 3.34 [95% CI: 2.93-3.81]; $P < 0.001$. Empa. = empagliflozin; Pbo. = placebo.

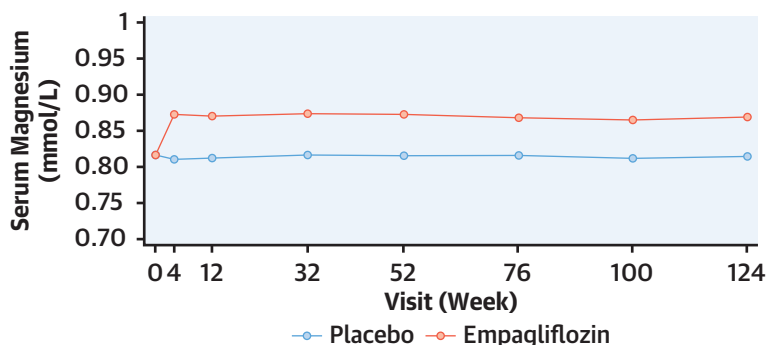
CENTRAL ILLUSTRATION Summary of the Study Main Findings

EMPEROR-Preserved (NCT03057951)

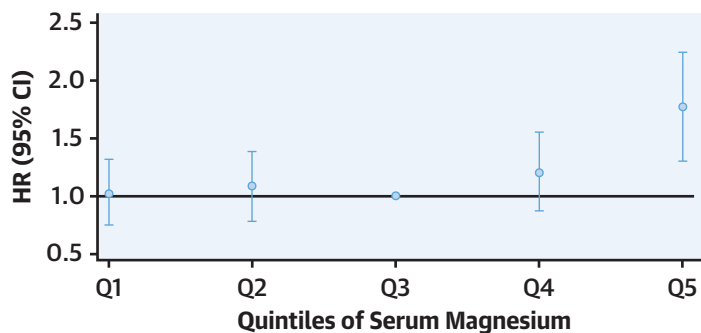
Chronic HF with mildly reduced or preserved ejection fraction

N = 5,988, randomized to either empagliflozin 10 mg/day or placebo

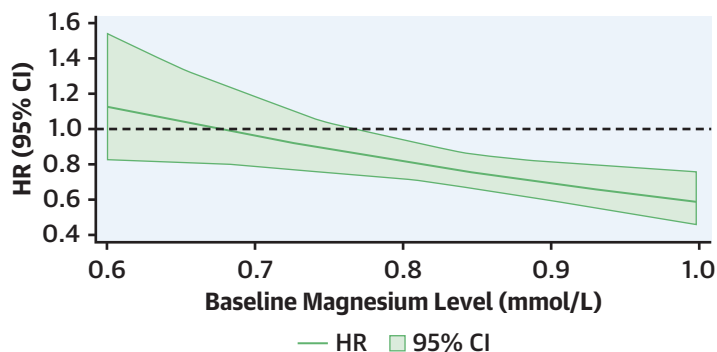
A Empagliflozin Increased Serum Magnesium Levels Over Time



B Higher Magnesium Levels Were Associated With HF Outcomes



C Empagliflozin Effect Was More Pronounced at Higher Magnesium



Ferreira JP, et al. JACC Heart Fail. 2026;14(2):102889.

(A) Empagliflozin (vs placebo) increased serum magnesium levels over time by 0.05 mmol/L. (B) Higher serum magnesium levels were associated with an increased risk of primary outcome events. (C) Empagliflozin (vs placebo) was associated with a more pronounced benefit on the primary outcome at higher serum magnesium levels. EMPEROR-Preserved = Empagliflozin Outcome Trial in Patients with Chronic Heart Failure with Preserved Ejection Fraction; HF = heart failure.

empagliflozin was greater at higher baseline magnesium levels. In HFrEF we also observed quite the opposite with a more pronounced benefit at lower baseline magnesium levels. This raises the hypothesis that correcting hypomagnesemia may be particularly relevant in HFrEF, in which chronic ATP depletion compromises myocardial energetics and magnesium is essential for ATP stabilization, mitochondrial function, and calcium-dependent processes underlying contractility.^{3,4,28} In contrast, because systolic function is relatively preserved in HFmrEF/HFpEF as in HFrEF, lower serum magnesium levels may have less impact on cardiac performance in these HF phenotypes. Furthermore, higher serum magnesium levels may stimulate autophagy by activating the adenosine-monophosphate activated-protein kinase (AMPK)/mammalian target of rapamycin (mTOR) pathway;²⁹ a pathway that has been described as central to the benefits of SGLT2 inhibitors, which benefits may be enhanced with higher magnesium levels in HFmrEF/HFpEF.³⁰ Irrespective of the several hypotheses, we refrained from performing a statistical “mediation” analysis because such analyses do not inform about the underlying mechanisms on how SGLT2 inhibitors increase serum magnesium nor how patients with HFmrEF/HFpEF and higher magnesium levels might have experienced a greater benefit from empagliflozin. In fact, such statistical “mediation” analysis may simply indicate an empagliflozin effect that influences the “mediator” (ie, serum magnesium) and that both the treatment and the “mediator” are associated with outcomes (which we know they are). Furthermore, in a post hoc analysis assessing the relation between serum magnesium levels and cardiovascular outcomes in the CANVAS (CANagliflozin cardioVascular Assessment Study) program, the authors demonstrated that post-randomization changes in serum magnesium were not a mediator of the effects of canagliflozin on cardiovascular outcomes.¹¹

SGLT2 inhibitors enhance the activity of the TRPM6 (Transient Receptor Potential cation channel subfamily M member 6) in both the kidney and the gut, and TRPM6 facilitates the reabsorption of magnesium in the colon and distal convoluted tubule explaining the rapid and sustained increase in serum magnesium levels with SGLT2 inhibitors.^{31,32} Importantly, TRPM6 activity is suppressed by oxidative stress and SGLT2 inhibitors may enhance TRPM6 activity by suppressing oxidative stress.³³ Therefore, beyond the aforementioned autophagy hypothesis, the larger magnitude of treatment effect among

patients with higher baseline magnesium levels also could be explained by an enhancement of TRPM6 activity and oxidative stress reduction with empagliflozin.³⁴ Furthermore, the higher risk of HF events associated with higher serum magnesium levels among these patients with HFmrEF/HFpEF also may have contributed to the greater treatment benefit observed in this subpopulation.

It has been documented that patients with T2D often present with low magnesium levels due to hypermagnesuria and insulin resistance.^{35–37} SGLT2 inhibitors correct hypomagnesemia in these patients through improved insulin sensitivity and reduced oxidative stress leading to restoration of TRPM6 activity.^{38,39} However, it is unclear why in the present study patients with lower magnesium levels at baseline appeared to experience an attenuated treatment benefit with empagliflozin regarding HF outcomes. It is possible that this represents a chance finding and further replication is warranted. In addition, the lower risk of events in this subgroup might also have mitigated the treatment effect. Of note, patients with hypertension presented more frequently with lower serum magnesium levels, a finding that may be confounded by a more frequent use of thiazide-type diuretic agents among patients with hypertension.

STUDY LIMITATIONS. This is a post hoc analysis of a randomized controlled trial and some of the findings might have occurred by chance, these include the magnesium-by-treatment interaction regarding the primary outcome, but do not include the effect of empagliflozin to increase serum magnesium levels that has been replicated with several SGLT2 inhibitors. Still, other studies with SGLT2 inhibitors in HFmrEF/HFpEF populations should try to replicate our findings. Second, we report associations between serum magnesium levels and outcomes, and we cannot make inferences about causality because such associations are prone to confounding. Third, we provided hypotheses on the mechanisms by which SGLT2 inhibitors increase serum magnesium levels; still, these need to be adequately tested. Last, magnesium is mostly an intracellular cation, and this study does not inform on intracellular magnesium levels.

CONCLUSIONS

In EMPEROR-Preserved including patients with HFmrEF/HFpEF, higher serum magnesium levels were associated with a higher risk of primary outcome events among patients treated with placebo but not among patients treated with empagliflozin

who experienced pronounced benefit. Compared with placebo, empagliflozin increased serum magnesium levels by 0.05 mmol/L in the overall population.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The sponsors of the trial were Boehringer Ingelheim and Eli Lilly and Company. Boehringer Ingelheim had organizational oversight over the EMPEROR trials, which included trial conduct, supervision and monitoring of the enrolling study centers, data collection and storage, as well as data analysis. The trial design was developed by the academic members of the Executive Committee in cooperation with representatives from Boehringer Ingelheim, who were also represented in the Executive Committee of the trial. The Executive Committee of the EMPEROR trials, consisting of academic members and representatives of Boehringer Ingelheim, developed the protocol and provided oversight of the trial's conduct together with the trial sponsor. The present analysis was performed through Vivli with the number 11010. This publication is based on research using data from data contributors Boehringer Ingelheim that has been made available through Vivli, Inc. Vivli has not contributed to or approved, and is not in any way responsible for, the contents of this publication. EMPEROR-Reduced (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction) was funded by Boehringer Ingelheim and Eli Lilly Company Diabetes Alliance. Dr Ferreira has received research support from Boehringer Ingelheim, AstraZeneca, Novartis, Bayer, Bial, and Salamandra. Dr Vasques-Nóvoa has received consulting or speaker fees from Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Daiichi Sankyo, Novartis, and Ultragenyx. All other authors have

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ADDRESS FOR CORRESPONDENCE: Prof João Pedro Ferreira, RISE-Health, Department of Surgery and Physiology, Faculty of Medicine, University of Porto, Alameda Prof. Hernâni Monteiro, 4200-319 Porto, Portugal. E-mail: jpferreira@med.up.pt.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Magnesium is an important cation for cardiac and vascular function. In HFmrEF/HFpEF, higher serum magnesium levels were associated with worse HF outcomes compared with normal magnesium levels. Empagliflozin increased serum magnesium levels and produced marked benefits among those with higher serum magnesium levels at baseline.

TRANSLATIONAL OUTLOOK: The role of magnesium in HFmrEF/HFpEF should be investigated further, particularly how the use of SGLT2 inhibitors may increase magnesium levels and how this influences the function of cardiomyocytes, endothelial cells, and autophagic flux.

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APPENDIX For supplemental tables, please see the online version of this paper.